

Top Trading Cycles in a practical setting

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June, 2026

Abstract

This thesis explores the Top Trading Cycles algorithm in the practical setting of the allocations of GPs, General Practitioner, in Norway for people wanting to switch or get a GP. It explores how different focus of *good* results can change both the runtime and results, and how to make the algorithm run as efficiently as possible. This thesis proposes an implementation of the Top Trading Cycles that can reduce the waitlists for GPs in Norway by over 80%.

Acknowledgements

I would like to thank my supervisor Fredrik Manne for his guidance and support throughout this project. I would also like to thank Juni Weisteen Bjerde for input during this project.

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1. Introduction

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1.1. Motivation

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1.2. Contributions

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1.3. Thesis Outline

The remainder of this thesis is organized as follows:

- **Section 2** provides background on the Top Trading Cycles mechanism and its application to GP allocation.
- **Section 3** formally defines the computational problems arising from TTC as variants of cycle cover problems on directed graphs.
- **Section 4** describes our Rust implementation and the algorithmic strategies employed.
- **Section 5** presents experimental results comparing different priority strategies.
- **Section 6** concludes and discusses future work.

2. Background

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2.1. The GP Allocation Problem in Norway

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2.2. Top Trading Cycles

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2.3. Related Work

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3. Problem Formalization

In this chapter we try to define the GP allocation problem and variations of it. In addition we go through how to construct a graph given a GP allocation problem and finally how we can reduce the GP allocation problem to the Cycle Cover problem in directed graphs.

3.1. GP allocation problem

3.1.1. Input

Consider the GP allocation problem as given a list of patients P and a list of doctors D . We then have patients $p_1, p_2, \dots, p_n$ and doctors $d_1, d_2, \dots, d_m$. We also are given the current and preferred doctor assignments as arrays:

- $D_{\text{current}}[i]$ denotes the index of the current doctor assigned to patient p_i (or $\perp$ if unassigned)
- $D_{\text{preferred}}[i]$ denotes the index of the preferred doctor for patient p_i

In addition we are given a priority function $R : P \rightarrow \mathbb{N}$, mapping some positive integer to each patient, the higher the number the higher the priority.

3.1.2. Feasible solution

A feasible solution for the GP allocation problem is a subset of patients $S \subseteq P$ such that the directed graph

$$G_S = (S, E_S), E_S = \{(a, b) \mid a, b \in S, D_{\text{preferred}}[\text{idx}(a)] = D_{\text{current}}[\text{idx}(b)]\} \quad (1)$$

consists of a vertex-disjoint union of directed cycles that cover all vertices in S . Where $\text{idx}(x)$ is a function giving the index of a patient x . So this means our solution S is a set of all patients that can exchange doctor in one or more cycles.

3.1.3. Optimization criterion

With this feasible solution defined we can start defining variants where we want feasible solution that maximizes some *score*. When choosing what patients should be exchanging we might have different qualities that we want in our solutions. We might want a solution that exchanges as many patients as possible or that exchanges patients with the highest priority.

First we define our general ordering

Definition 3.1.3.1 (Optimization criterion): An ordering $\succ$ over feasible solutions. A solution is optimal if it is maximal under $\succ$.

Then we can build upon this to create our two variants

3.1.3.1. Lexicographic maximization by priority

Definition 3.1.3.1.1 (Characteristic vector): Let patients be ordered by priority: $p^{(1)} \succ p^{(2)} \succ \dots \succ p^{(n)}$ where $p^{(1)}$ has the highest priority. Given a feasible solution $S \subseteq P$, the **characteristic vector** is:

$$\chi(S) = (b_1, b_2, \dots, b_n) \in \{0, 1\}^n, \quad b_i = \begin{cases} 1 & \text{if } p^{(i)} \in S \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Definition 3.1.3.1.2 (Lexicographic ordering by priority):

$$S \succ_{\text{lex}} S' \quad \text{iff} \quad \chi(S) \text{ is lexicographically greater than } \chi(S') \quad (3)$$

That is, at the first index i where they differ, $\chi(S)_i = 1$ and $\chi(S')_i = 0$.

Intuitively, $\chi(S)$ is a binary number whose most significant bit corresponds to the highest-priority patient. The optimal solution is the one that maximises this binary number: always satisfying the highest-priority patient it can, then subject to that the next highest, and so on.

Example (Ordering two solutions lexicographically by priority):

We use the example graph G below as the graph to create solutions from.

Given solutions

1. $S = \{P_4, P_5\}$ represents the subgraph G_S containing cycle $P_4 \rightarrow P_5 \rightarrow P_4$
2. $S' = \{P_1, P_2, P_3\}$ represents the subgraph $G_{S'}$ containing cycle $P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_1$

For simplicity's sake we say that $R(P_i) = i$, so P_5 has the highest priority.

Ordering all five patients by priority gives $p^{(1)} = P_5, p^{(2)} = P_4, \dots, p^{(5)} = P_1$. Then:

$$\chi(S) = (1, 1, 0, 0, 0) \quad \chi(S') = (0, 0, 1, 1, 1) \quad (4)$$

At the first differing position ($i = 1$), $\chi(S)_1 = 1 > 0 = \chi(S')_1$, so $S \succ_{\text{lex}} S'$.

So for our first variant we want to find the maximal solution under the $\succ_{\text{lex}}$ ordering. This means always prioritizing those with highest priority.

3.1.3.2. Maximizing cardinality

The other variant is finding a solution with the largest cardinality, e.g. the solution that contains the most patients.

Definition 3.1.3.2.1 (Ordering by cardinality):

$$S \succ_{\text{size}} S' \quad \text{iff} \quad |S| > |S'| \quad (5)$$

This solution will be the one that exchanges the most patients and therefore makes the most amount of people happy.

Example (Ordering two solutions by cardinality):

Using the same solutions as the previous example:

1. $S = \{P_4, P_5\}$ represents the subgraph G_S containing cycle $P_4 \rightarrow P_5 \rightarrow P_4$
2. $S' = \{P_1, P_2, P_3\}$ represents the subgraph $G_{\{S'\}}$ containing cycle $P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_1$

Under $\succ_{\text{size}}: |S| = 2$ and $|S'| = 3$, so $S' \succ_{\text{size}} S$. The absolute maximal solution is $\{P_1, P_2, P_3, P_4, P_5\}$, the union of both cycles.

Notice that the same two sets are ordered **oppositely** under the two criteria: $S \succ_{\text{lex}} S'$ (because S contains the highest-priority patient P_5) but $S' \succ_{\text{size}} S$ (because S' has more patients).

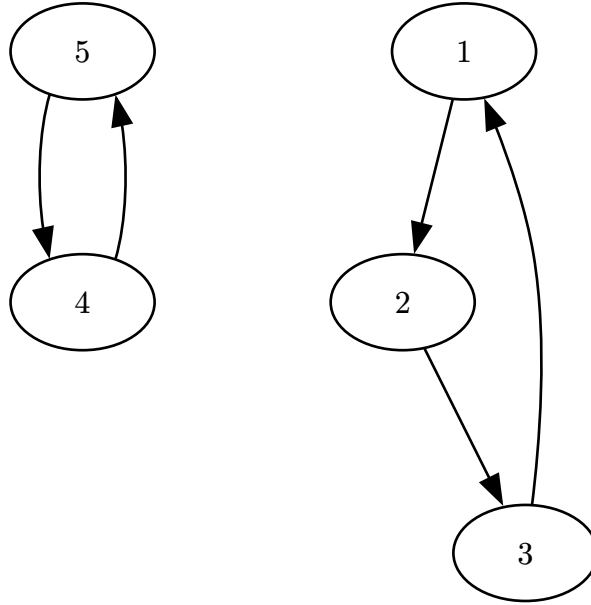


Figure 1: An example graph G .

4. Implementation

4.1. Greedy DFS

We first consider a greedy approach inspired by the Top Trading Cycles implementation of Huitfeldt et al. Their algorithm preserves Pareto optimality at each round by restricting each doctor node to a single outgoing edge. We relaxed this constraint, allowing each doctor to maintain outgoing edges to all of its current patients simultaneously, and resolved ties by patient priority.

We are given the lists of patients and doctors P, D , the assignment arrays $D_{\text{cur}}, D_{\text{pref}}$, and a priority function $R : P \rightarrow \mathbb{N}$, where a higher number means higher priority. We model the problem as a bipartite directed graph over patients and doctors. Let $I = \{0, \dots, |P| - 1\}$. Then:

$$G = (V, E), \quad V = P \cup D, \quad E = \left\{ (p_i, D_{\text{pref}[i]}) \mid i \in I \right\} \cup \left\{ (D_{\text{cur}[i]}, p_i) \mid i \in I \right\}$$

Each patient p_i has an outgoing edge to their preferred doctor, and each doctor has outgoing edges to all of its currently registered patients. A directed cycle in G necessarily alternates between patient and doctor nodes and corresponds to a valid exchange: each patient in the cycle moves to their preferred doctor, and each doctor loses exactly one patient while gaining one.

The algorithm processes patients in decreasing priority order. For each patient, a DFS attempts to find a cycle through that patient; when the DFS reaches a doctor node with multiple candidate patients, it always explores the highest-priority one first.

```
1 let resolved_patients = []
2 let p_prio = Sorted list of patients by priority
3 let wants_to_switch = [true] * |P|
4 for each  $p \in \text{p\_prio}$  do
5   if let cycle = dfs_find_cycle(G, R, p) then
6     for each  $p \in \text{cycle}$  do
7       let  $i = \text{idx}(p)$ 
8       wants_to_switch[i] = false
9       resolved_patients.push(p)
10    end
11  end
12 end
13 return resolved_patients
```

The greedy rule ensures that high-priority patients are preferentially included in cycles, but it does not guarantee a globally optimal solution. The following example illustrates how the greedy choice can be locally motivated yet globally suboptimal.

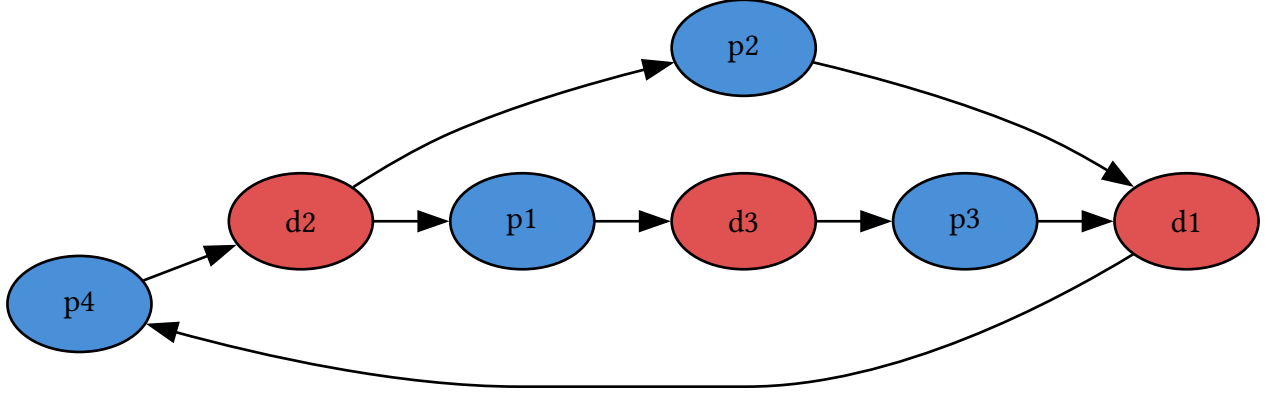


Figure 2: Example graph G where algorithm would choose suboptimal solution. px means patient x and has priority x .

In Figure 2 each patient px has priority x . The DFS begins at $p4$ (highest priority), follows the edge to $d2$, and there chooses $p2$ over $p1$ since $R(p2) > R(p1)$. The resulting cycle $p4 \rightarrow d2 \rightarrow p2 \rightarrow d1 \rightarrow p4$ is committed, leaving $p1$ and $p3$ unsatisfied with no further cycles remaining.

Had the DFS chosen $p1$ at $d2$ instead, it would have found the longer cycle $p4 \rightarrow d2 \rightarrow p1 \rightarrow d3 \rightarrow p3 \rightarrow d1 \rightarrow p4$, satisfying three patients. The greedy choice at $d2$ was locally motivated by priority but globally suboptimal. This motivates the exact algorithms in the following sections.

4.2. Exact algorithm for maximizing total switches

This algorithm applies the classical cycle canceling technique from network flow theory to the GP-switching problem, finding the maximum-cardinality feasible solution in polynomial time. The key insight is that the problem reduces to a maximum integer circulation, for which efficient algorithms are known.

4.2.1. Graph structure

Rather than working with the bipartite patient-doctor graph, we compress to a weighted directed graph over doctors only. Patients who want the same switch are interchangeable: all that matters is how many can be routed. We therefore aggregate them into edge weights:

$$\begin{aligned}
 G &= (V, E), \quad V = D \\
 E &= \left\{ (D_{\text{cur}[i]}, D_{\text{pref}[i]}) \mid i \in \llbracket P \rrbracket \right\} \\
 w(a, b) &= |\{i \in \llbracket P \rrbracket \mid D_{\text{cur}[i]} = a \wedge D_{\text{pref}[i]} = b\}|
 \end{aligned} \tag{6}$$

A cycle $d_1 \rightarrow d_2 \rightarrow \dots \rightarrow d_k \rightarrow d_1$ carrying f units of flow corresponds to f patients rotating around the cycle: each moves from their current doctor to the next in the cycle. This compression reduces the graph from $O(|P| + |D|)$ nodes to $O(|D|)$ nodes, which is significant when many patients share the same switch request.

Applying this transformation to Figure 2 gives the doctor graph in Figure 3, where we can still identify the short cycle $d1 \rightarrow d2$ and the longer cycle $d1 \rightarrow d2 \rightarrow d3$.

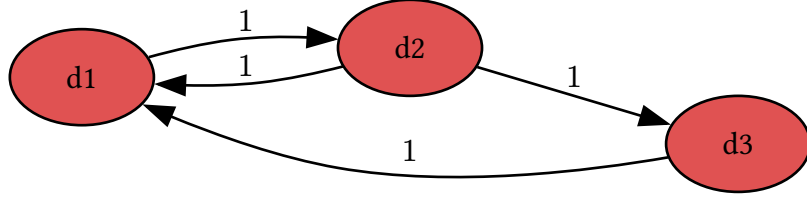


Figure 3: Graph from Figure 2 but in doctor graph structure.

4.2.2. Algorithm

We want to find, for each edge $e \in E$, a non-negative integer $f(e)$ representing how many of the $w(e)$ patients on that edge actually switch. Two conditions make such an assignment a valid set of simultaneous exchanges:

- **Capacity:** $0 \leq f(e) \leq w(e)$ for all $e \in E$
- **Flow conservation:** for every doctor $d \in D$: $\sum_{(v,d) \in E} f(v,d) = \sum_{(d,v) \in E} f(d,v)$

Flow conservation is exactly what forces the assignment to decompose into cycles: every doctor who loses a patient must gain one. The problem is therefore:

Problem: Maximum Switch Circulation

Input: A directed graph $G = (D, E, w)$ where $w : E \rightarrow \mathbb{N}$

Output: An integer function $f : E \rightarrow \mathbb{N}_0$ maximising $\sum_{e \in E} f(e)$ subject to:

- $0 \leq f(e) \leq w(e)$ for all $e \in E$
- $\sum_{(v,u) \in E} f(v,u) = \sum_{(u,v) \in E} f(u,v)$ for all $u \in D$

This is a **maximum integer circulation** problem, solvable in polynomial time. We solve it using the **successive positive cycle** method.

Given a current flow f , the **residual graph** G_f is built by replacing each original edge (u, v) with:

- a **forward residual** arc (u, v) with capacity $w(u, v) - f(u, v)$ and cost $+1$
- a **backward residual** arc (v, u) with capacity $f(u, v)$ and cost -1

A directed cycle in G_f is **positive** if the sum of its arc costs is positive, meaning it contains more forward than backward arcs, so augmenting along it strictly increases total flow. The optimality condition for maximum circulations states that f is optimal if and only if G_f contains no positive cycle.

The outer loop finds and augments positive cycles until none remain:

```

1 for each  $e \in E$  do  $f(e) \leftarrow 0$  end
2 while FindPositiveCycle( $G_f$ ) returns a cycle  $C$  do
3    $b \leftarrow \min_{e \in C} r_f(e)$   $\triangleright$  bottleneck residual capacity
4   for each  $(u, v) \in C$  do

```

```

5 | |  $f(u, v) \leftarrow f(u, v) + b$ ; update residual capacities in  $G_f$ 
6 | end
7 end
8 return  $f$ 

```

Positive cycles are detected using a variant of Bellman–Ford (SPFA). All nodes are seeded with distance 0, simulating a virtual super-source at zero cost so every cycle is reachable. Edges are relaxed greedily in the maximising direction; a node relaxed $|D|$ or more times must lie on a positive cycle.

```

1 for each  $v \in D$  do  $\text{dist}[v] \leftarrow 0$ ;  $\text{pred}[v] \leftarrow \perp$ ; enqueue  $v$  end
2 while queue not empty do
3 |  $u \leftarrow \text{dequeue}$ 
4 | for each  $(u, v)$  in  $G_f$  with  $r_f(u, v) > 0$  do
5 | | if  $\text{dist}[u] + 1 > \text{dist}[v]$  then
6 | | |  $\text{dist}[v] \leftarrow \text{dist}[u] + 1$ ;  $\text{pred}[v] \leftarrow u$ 
7 | | | if  $v$  has been enqueued  $\geq |D|$  times then
8 | | | | Walk back  $|D|$  steps from  $v$  along  $\text{pred}$  to find node  $c$  on the cycle
9 | | | | Collect and return the cycle starting and ending at  $c$ 
10 | | | end
11 | | | Enqueue  $v$  if not already in queue
12 | | end
13 | end
14 end
15 return  $\emptyset$   $\triangleright$  no positive cycle;  $f$  is optimal

```

Theorem 4.2.2.1 (Optimality of MaxSwitchCirculation): The algorithm returns a flow f achieving the maximum possible value $\sum_{e \in E} f(e)$, equal to the maximum number of patients that can simultaneously switch to their preferred doctor.

Theorem 5: Optimality of MaxSwitchCirculation

Proof: The algorithm terminates when FindPositiveCycle returns $\emptyset$, meaning G_f contains no positive-cost directed cycle. By the cycle optimality criterion for maximum circulations: a feasible integer circulation f is optimal if and only if its residual graph G_f contains no positive-cost cycle. Since each forward arc carries cost $+1$ and each backward arc cost -1 , any positive-cost cycle in G_f would strictly increase $\sum f(e)$; the absence of such cycles certifies that f is maximum. $\square$

Termination and complexity. Each call to FindPositiveCycle runs SPFA over $|D|$ nodes and at most $|D|^2$ edges in $O(|D|^3)$ time. Each augmentation round increases total flow by

at least 1, so the number of rounds is at most $W = \sum_{e \in E} w(e)$, the total number of patients wanting to switch, giving a worst-case bound of $O(W \cdot |D|^3)$. In practice each round pushes a bottleneck of $b \geq 1$ units, so the number of distinct rounds is bounded by the number of edges that become saturated, at most $|E| \leq |D|^2$. This gives the graph-size bound $O(|D|^5)$, independent of the patient count.

4.3. Exact algorithm for maximizing priority

This algorithm finds the lexicographically maximal feasible solution under $\succ_{\text{lex}}$, as defined in Section 3. It builds on the same residual-graph framework but processes patients in priority order, greedily committing each one as soon as a cycle is found. The correctness of this greedy strategy follows from the theorem below.

Theorem 4.3.1 (Greedy choice property): Let p^* be the highest-priority patient that belongs to some feasible solution. Then every lexicographically maximal solution contains p^* .

Theorem 6: Greedy choice property

Proof: Suppose S^* is a lexicographically maximal feasible solution with $p^* \notin S^*$. By hypothesis there exists a feasible solution S' with $p^* \in S'$. Let k be the index of p^* in the priority ordering $p^{(1)} \succ p^{(2)} \succ \dots \succ p^{(n)}$. Every patient with index $i < k$ belongs to no feasible solution by the choice of p^* , so $\chi(S^*)_i = \chi(S')_i = 0$ for all $i < k$. At index k : $\chi(S')_k = 1 > 0 = \chi(S^*)_k$. Therefore $S' \succ_{\text{lex}} S^*$, contradicting the maximality of S^* . $\square$

The algorithm applies this observation iteratively: process patients from highest to lowest priority and commit each patient to a cycle as soon as one is found.

We use the same doctor graph as before: doctors are nodes, and each patient p with current doctor u and preferred doctor v is a directed arc $a_p = u \rightarrow v$ with capacity 1. Each arc also has a corresponding **backward arc** $\bar{a}_p = v \rightarrow u$ with initial capacity 0.

Pruning. Before processing, we compute the strongly connected components (SCCs) of the graph. Any patient whose current and preferred doctor lie in different SCCs, or in a trivial SCC of size 1, can never be part of any directed cycle and is immediately discarded. This avoids unnecessary BFS calls and keeps the residual graph clean throughout execution.

The algorithm processes the remaining patients from highest to lowest priority. For each patient p (arc $u \rightarrow v$), exactly one of two cases applies:

- **Case 1 (primary arc available, $\text{cap}(a_p) > 0$):** p has not yet been committed. We run BFS from v to u in the current residual graph. If a path Π exists, then Π together with a_p forms a cycle, and we commit p :
 - The primary arc a_p is consumed **permanently**: its backward arc capacity stays 0, so no future patient can traverse it in reverse and undo p 's switch.
 - Each routing arc a in Π is consumed normally: $\text{cap}(a)$ decreases by 1 and $\text{cap}(\bar{a})$ increases by 1, leaving a residual that lower-priority patients may use to reroute.

- If no path exists, p is skipped.
- **Case 2 (primary arc consumed, residual active, $\text{cap}(a_p) = 0$ and $\text{cap}(\overline{a_p}) > 0$):** p 's arc was already consumed as a **routing arc** in an earlier (higher-priority) patient's cycle, which created the residual $\overline{a_p}$. We now **solidify** by setting $\text{cap}(\overline{a_p}) \leftarrow 0$, preventing any lower-priority patient from traversing this residual to undo p 's assignment.

```

1  Compute SCCs of  $G$ ; discard patients whose arc spans different or trivial SCCs
2   $E_{\text{sort}} \leftarrow$  remaining patients sorted by priority descending
3  for each patient  $p$  with arc  $a_p = (u \rightarrow v)$  in  $E_{\text{sort}}$  do
4      if  $\text{cap}(a_p) > 0$  then
5          if  $\text{BfsPath}(v, u)$  in residual graph returns path  $\Pi$  then
6               $\text{cap}(a_p) \leftarrow 0$  ▷ consume primary permanently; no residual
7              for each arc  $a \in \Pi$  do
8                   $\text{cap}(a) \leftarrow \text{cap}(a) - 1$ ;  $\text{cap}(\overline{a}) \leftarrow \text{cap}(\overline{a}) + 1$ 
9              end
10             Add  $p$  to  $S$ 
11         end
12     else if  $\text{cap}(\overline{a_p}) > 0$  then
13          $\text{cap}(\overline{a_p}) \leftarrow 0$  ▷ solidify; lock in routing commitment
14         Add  $p$  to  $S$ 
15     end
16 end
17 return  $S$ 

```

Why the primary arc has no residual: This is the core invariant. Once patient p is committed as the beneficiary of a cycle, no future patient should be able to undo it. By not creating a residual for a_p , no BFS can ever traverse backwards through p 's switch.

Why routing arcs do have residuals: A lower-priority patient may be able to “take over” a routing role. For example, if p_1 (high priority) uses p_3 's arc as routing, a later patient p_2 may form a cycle that routes through p_3 's arc differently, replacing it. This is valid: only the final destination of p_1 (whose primary arc is irrevocable) is fixed.

Theorem 4.3.2 (Correctness): The algorithm returns the lexicographically maximal feasible solution.

Theorem 7: Correctness

Proof: By induction on the priority ordering. The Greedy Choice Theorem establishes the base case: the highest-priority feasible patient p^* must appear in every lex-max solution, and BFS correctly identifies whether p^* is feasible, since a path from v to u exists in G if and only if p^* lies on some directed cycle.

For the inductive step, assume all higher-priority patients have been correctly committed. The residual graph reflects these commitments: primary arcs are permanently consumed and cannot be undone, while routing residuals remain available. We claim a path from v to u exists in the current residual graph if and only if p can be added to a feasible solution extending all committed patients. This follows from the augmenting-path argument: any feasible cycle through p either avoids all committed arcs and is directly reachable in G_f , or shares some routing arcs whose residuals permit an equivalent rerouting. The solidify step in Case 2 maintains the invariant: once a patient's arc is confirmed as routing for a higher-priority cycle, its residual is removed so no lower-priority patient can undo the commitment. $\square$

Complexity. Sorting patients requires $O(|P| \log |P|)$. The initial SCC computation runs in $O(|D| + |E|)$. Each patient requires at most one BFS over the residual graph in $O(|D| + |E|)$ time. Processing all $|P|$ patients therefore takes $O(|P| \cdot (|D| + |E|))$. Since $|E| \leq |D|^2$, the overall complexity is $O(|P| \cdot |D|^2)$.

4.4. Metaheuristics

5. Experimental Results

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5.1. Experimental Setup

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5.2. Performance Comparison

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5.3. Impact of Priority Strategies

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6. Conclusion

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6.1. Summary

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6.2. Future Work

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